SPECTROSCOPIC STUDIES OF PRISTINE LUNAR REGOLITH UNDER H_2O, O_2, AND CO_2 CONTROLLED CONDITIONS

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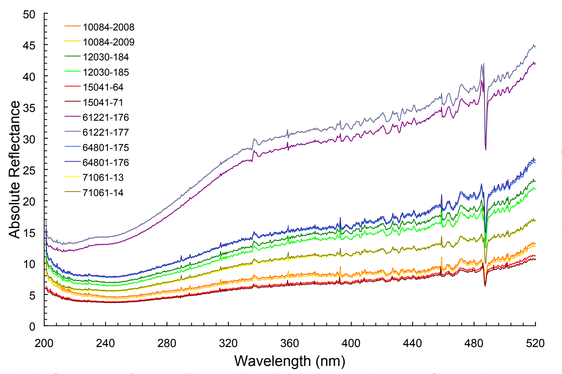
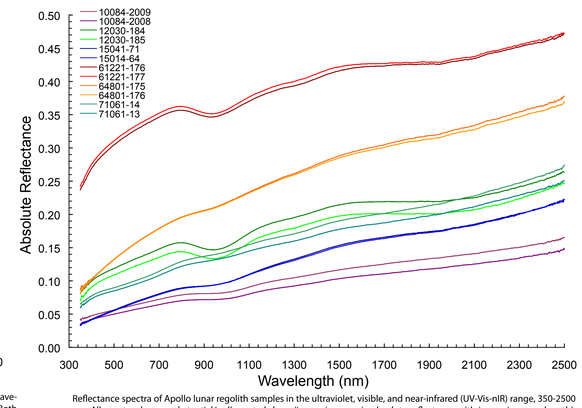
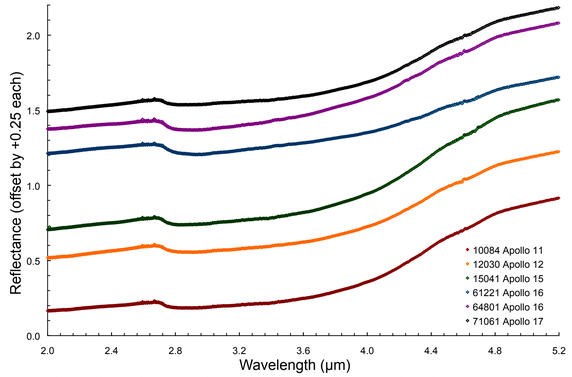
Lunar materials are highly susceptible to contamination by Earth's atmosphere and hydrosphere. Many spectroscopic features of lunar regolith that are of high interest, including the effects of nanoscopic Fe^0 particles generated by space weathering processes, and the possible presence of surface hydroxyl and adsorbed molecular water cannot easily be studied in terrestrial laboratories. Further, even trace levels of Fe^{3+} produced by terrestrial contamination can significantly alter the spectral properties of lunar materials. In order to provide improved ground-truth for remote sensing data interpretation, we have implemented a range of contamination controls in order to carry out a reflectance spectroscopic study of lunar regolith samples that have been curated in a pristine state since collection and return to Earth by the Apollo astronauts.



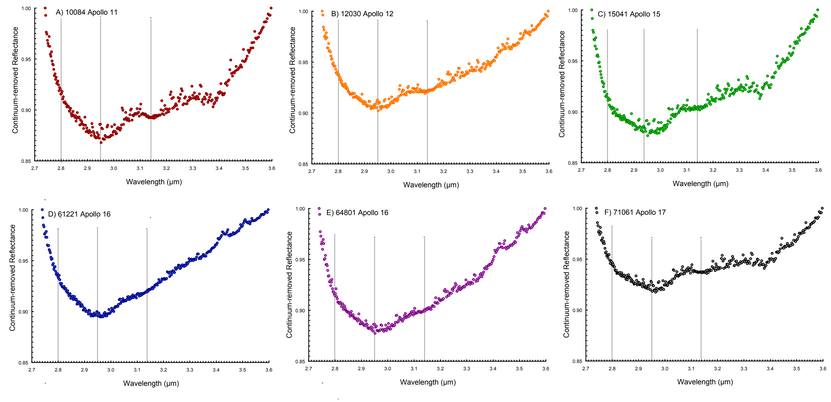
Ultraviolet reflectance spectra of pristine Apollo lunar regolith (A). All spectra show a general increase in reflectance with increasing wavelength. Continuum-removal reveals a broad band centred near ~260 nm that is ascribable to $^{VI}Fe^{2+}$, with a lesser contribution from $^{VI}Ti^{3+}$. Both $^{VI}Fe^{2+}$ and $^{VI}Ti^{3+}$ occur in a wide variety of specific sites, including the M1 and M2 sites in pyroxene, the M site of plagioclase, octahedral sites in ilmenite and other Fe-Ti oxides, and octahedral sites in silicate glasses; this produces a correspondingly broad absorption feature. A weaker feature near ~216 nm is tentatively ascribed to $^{VI}Ti^{3+}$ the host phase(s) of which are presently unknown. Small, sharp features are due to imperfect correction for D_2 molecular lines in spectrum of the deuterium lamp light source. Continua were removed by division by straight line segments, the first anchored at 202 nm and 229 nm, the second at 229 nm and 375 nm. Feature centre locations were determined by fitting 3rd-order polynomials to the continuum-removed spectra.



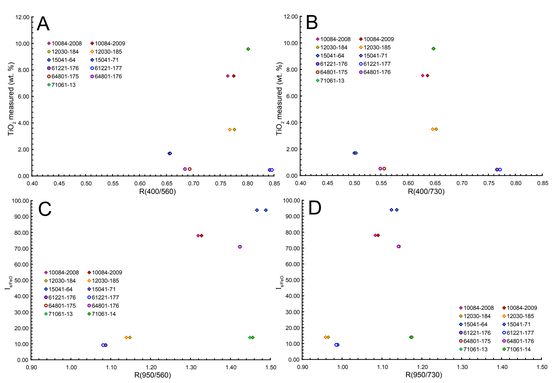
Reflectance spectra of Apollo lunar regolith samples in the ultraviolet, visible, and near-infrared (UV-Vis-nIR) range, 350-2500 nm. All spectra show a substantial 'red' spectral slope (i.e., an increase in absolute reflectance with increasing wavelength). Various spectral features are discernable. Bands centred near ~950 nm and ~2000 nm are dominantly ascribable to Fe^{2+} in the octahedral sites of silicate minerals and glasses. The ~950 nm feature is due primarily to the less-distorted M1 site in pyroxenes, and the ~2000 nm feature to the more highly distorted M2 site in pyroxenes. A weaker feature centred near 1250 nm is due primarily to trace levels of Fe^{2+} substituting for Ca^{2+} in the highly-distorted M site of plagioclase. The Apollo 17 samples 71061-14 and 71061-13, and the Apollo 12 samples 12030-184 and 12030-185 show a broad, shallow feature centred near 600 nm that is ascribable to a combination of Fe^{2+}-Ti^{4+} charge transfer features in ilmenite, mafic glasses, and other (Fe,Ti)-bearing phases, possibly with a contribution from Ti^{3+}.



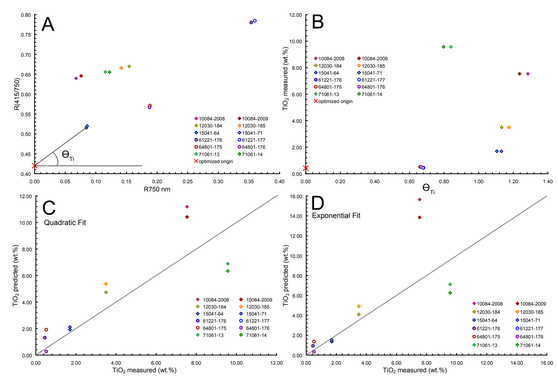
Reflectance spectra of pristine lunar regolith samples from 2.0 to 5.2 μm showing absorption features due to OH and H2O. The "3-micron" feature is a broad envelope ranging from ~2.70 μm to ~4.45 μm wavelength. This feature is a mixture of numerous O-H and H-O-H vibrational overtones and combinations, which are poorly resolved because of the vast diversity of possible O-H co-ordination environments in surficial and structural OH and adsorbed H_2O. The weak feature between ~4.5 μm and ~4.8 μm is due to the first overtones of the silicate Si-O fundamental vibrations. A) Spectra in absolute reflectance, shifted vertically by +0.25 each for clarity. B) Spectra normalized to the reflectance averaged over all bands between 2617 nm and 2697 nm (2.617-2.697 μm; the range over which the approximate continuum were computed for M3 spectra).



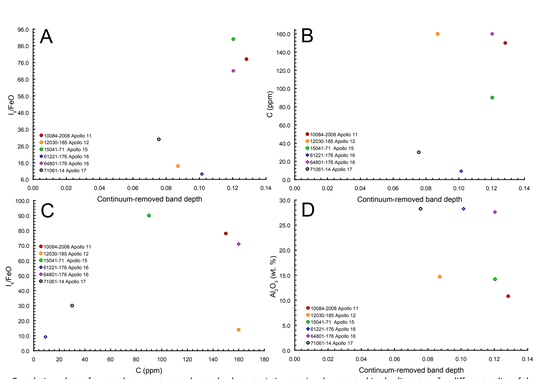
Continuum-removed spectra of lunar regolith samples reveal details of band structure. These spectra are similar to those measured by Cassini VIMS (cf. Fig 2B of Clark, 2009) and Deep Impact/EPOXI (cf. Fig. 1D and 4A-D of Sunshine et al., 2009). All spectra show an inflection near 2.80 μm and local minima near 2.96 μm and 3.14 μm (marked by thin vertical lines), which are similar values to those in continuum-removed spectra of the lunar surface measured by Deep Impact/EPOXI (2.81 μm, 2.95 μm and 3.14 μm).



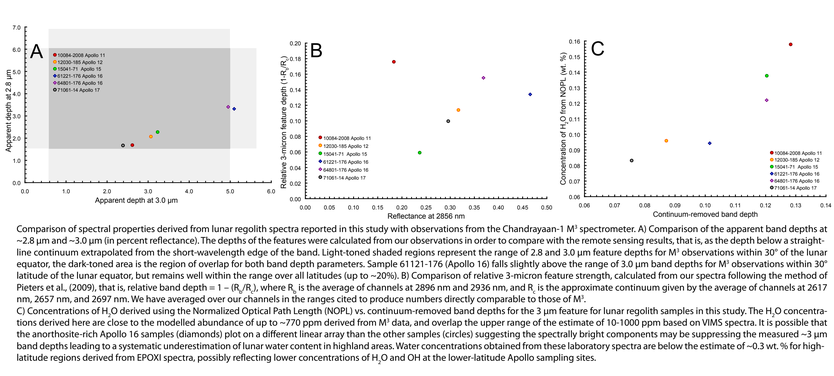
Band ratio plots developed for remote determination of lunar surface TiO_2 concentration from telescopic observations. A) Charette et al., (1974) developed a calibration curve for TiO_2 and ratio plots developed for remote determination of lunar surface TiO_2 concentration from telescopic observations. A) Charette et al., (1974) developed a calibration curve for TiO_2 concentration based on the 400 nm to 560 nm reflectance ratio. Johnson et al., (1991a) demonstrated that additional compositional contrast could be attained in lunar surface maps by using the 400 nm to 730 nm reflectance ratio (B). Plots of both 400/560 and 400/730 nm band ratio vs. TiO_2 concentration show significant scatter, likely because our samples include a wide range of soil maturities whereas the methods developed by Charette et al., (1974) and modified by Johnson et al., (1991a) used only mature to submature soils. C) McCord et al., (1976) showed that the 950 nm to 560 nm reflectance ratio provides a useful measure of lunar surface maturity. This technique was subsequently modified by Larson et al., (1991) using the 950 nm to 730 nm reflectance ratio (D). Comparison of both the 950 nm to 730 nm and 950 nm to 560 nm reflectance ratios with to the ferromagnetic resonance surface maturity index $I_{s/FeO}$ reveals that both spectral ratios show a reasonable correlation with the $I_{s/FeO}$ parameter apart from Apollo 17 sample 71061, due to the effect of the ~600 nm feature. Apart from the anomalous sample 71061, there appears to be a reasonable correlation between both 950/560 nm and 950/730 nm spectral ratios and $I_{s/FeO}$ values.



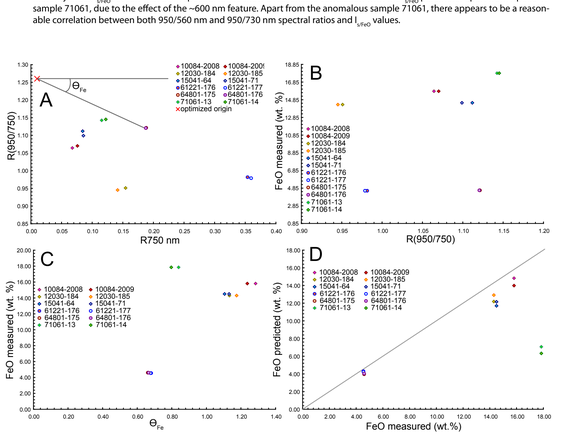
Spectral-compositional relations related to TiO_2 following methods developed for the Clementine global multispectral data set. A) The parameter Θ_{Ti} is an angular parameter designed to minimize the effects of surface maturity. B) The relation between Θ_{Ti} and TiO_2 concentrations is clearly non-linear, and Apollo 17 sample 71061 is anomalous, due primarily to the ~600 nm band. Nevertheless, empirical relationships developed between Θ_{Ti} and TiO_2 concentrations produce reasonable estimates, with larger misfits between measured and calculated values at higher TiO_2 concentrations. Both quadratic (C) and exponential (D) fits produce reasonable results, with somewhat more scatter in the exponential fit, however, this must be interpreted in light of the small number of samples measured in this study. Again note the anomalous behaviour of sample 71061.



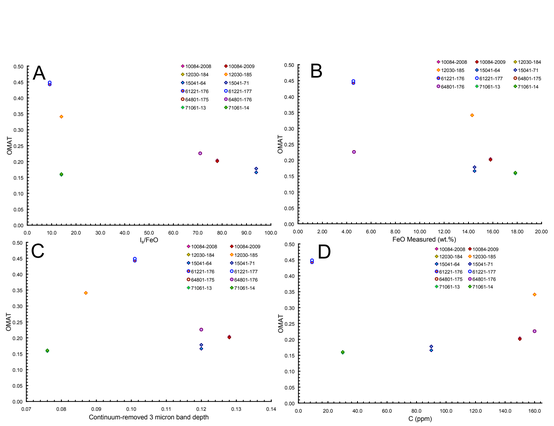
Correlation plots of spectral parameters and sample characteristics previously reported in the literature for different splits of the same parentage as the samples studied here. A) Continuum-removed band depth of the ~3 μm feature in Apollo lunar samples is correlated with lunar soil maturity parameter $I_{s/FeO}$. The soil maturity parameter is the ratio between ferromagnetic resonance intensity at g ~2.1 (g is the dimensionless magnetic moment) to the FeO concentration. The relationship between increased hydration feature depth and maturity parameter supports solar wind H implantation as the predominant source of lunar surface OH and H_2O. B) The general correlation between C concentration and continuum-removed ~3 μm feature depth may reflect input from solar wind implantation and a variety of infalling matter including cometary material, carbonaceous chondrites and interplanetary dust particles, therefore, Apollo 12 sample 12030 appears anomalously rich in C compared to the continuum-removed band depth, indicating that the addition of carbon to the lunar regolith is not always accompanied by hydration. C) There is a general correlation between soil maturity parameter and C concentration for these samples, but the relationship between C concentration, maturity and hydration is not simple, possibly due to overlapping or competing processes involved (solar wind implantation, infall of exogenous matter, and both creation and destruction/removal of OH/H_2O). Sample 12030 is again anomalous, with a very high C concentration for its maturity. D) Although some correspondence between highland material rich in Al (primarily due to anorthitic plagioclase) and hydration spectral features has been noted, the lack of a strong correlation between continuum-removed ~3 μm feature band depth and Al2O3 content in these soil samples may indicate that a compositional or mineralogical control on OH/H2O generation is not the predominant mechanism responsible for the observed 3 μm feature enhancement in lunar spectra.



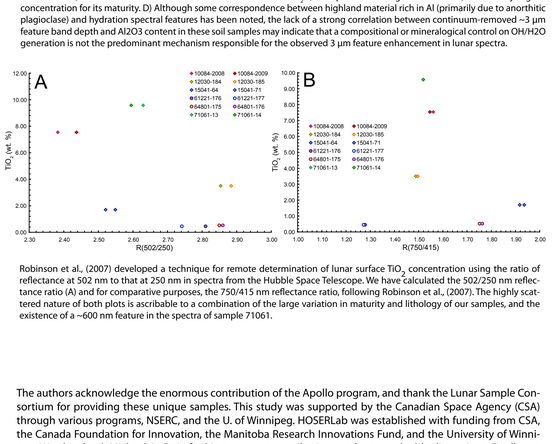
Comparison of spectral properties derived from lunar regolith spectra reported in this study with observations from the Chandrayaan-1 M^3 spectrometer. A) Comparison of the apparent band depths at ~2.8 μm and ~3.0 μm (in percent reflectance). The depths of the features were calculated from our observations in order to compare with the remote sensing results, that is, as the depth below a straight-line continuum extrapolated from the short-wavelength edge of the band. Light-toned shaded regions represent the range of 2.8 and 3.0 μm feature depths for M^3 observations within 30° of the lunar equator, the dark-toned area is the region of overlap for both band depth parameters. Sample 61121-176 (Apollo 16) falls slightly above the range of 3.0 μm band depths for M^3 observations within 30° latitude of the lunar equator, but remains well within the range over all latitudes (up to ~20%). B) Comparison of relative 3-micron feature strength, calculated from our spectra following the method of Pieters et al., (2009), that is, relative band depth = 1 – (R_c/R_s), where R_c is the average of channels at 2896 nm and 2936 nm, and R_s is the approximate continuum given by the average of channels at 2617 nm, 2657 nm, and 2697 nm. We have averaged over our channels in the ranges cited to produce numbers directly comparable to those of M^3.
C) Concentrations of H_2O derived using the Normalized Optical Path Length (NOPL) vs. continuum-removed band depths for the 3 μm feature for lunar regolith samples in this study. The H_2O concentrations derived here are close to the modelled abundance of up to ~770 ppm derived from M^3 data, and overlap the upper range of the estimate of 10-1000 ppm based on VIMS spectra. It is possible that the anorthosite-rich Apollo 16 samples (diamonds) plot on a different linear array than the other samples (circles) suggesting the spectrally bright components may be suppressing the measured ~3 μm band depths leading to a systematic underestimation of lunar water content in highland areas. Water concentrations obtained from these laboratory spectra are below the estimate of ~0.3 wt. % for high-latitude regions derived from EPOXI spectra, possibly reflecting lower concentrations of H_2O and OH at the lower-latitude Apollo sampling sites.



Parameters related to FeO concentration in lunar surface materials derived from spectral data following methods developed for the Clementine global multispectral data set. A) The plot of the reflectance ratio of the 950 nm and 750 nm channels versus reflectance at 750 nm shows the complex relationship between FeO, surface maturity and spectral properties. The Euclidian distance between a point on this plot and the origin is correlated with surface maturity (space weathering, in particular the presence of nanoscopic Fe0 particles), while the angle between the vector from a point to the optimized origin and the horizontal is a measure of FeO content. B) There is no clear correlation between the simple spectral ratio and measured FeO concentration, due primarily to the effects of variable surface maturity; it is these maturity effects that the methods developed by P.G. Lucey and co-workers were designed to minimize. C) There is a good linear correlation between the angular parameter Θ_{Fe} and FeO content, with the exception of Apollo 17 sample 71061, which deviates from the best-fit line through the other points. The mismatch in the case of 71061 is due mostly to the broad shallow feature centred near 600 nm, which causes changes in the intensity at 750 nm. FeO concentrations predicted using these methods are close to the concentrations measured on splits of the same materials except again for sample 71061. In all cases, however, the predicted FeO content is slightly below the measured value. Possible explanations include incorporation of Fe into materials which do no strongly influence the reflectance values at 750 nm 950 nm, (e.g., plagioclase, ilmenite, Fe0, Fe sulphides, some glasses); or non-FeO components influencing the 750 nm 950 nm intensities (e.g., other transition metal cations in octahedral sites).



Optical maturity parameter (OMAT) is a parameter developed by P. G. Lucey at co-workers for measuring the maturity of the lunar surface imaged by Clementine. The OMAT parameter is the Euclidian distance between the point on a plot of the ratio of reflectance at 950 nm to that at 750 nm versus reflectance at 750 nm and the optimized origin corresponding to a 'hypermature' end-member; note that therefore the value of OMAT approaches zero as maturity increases. A) We have computed the OMAT values for our suite of pristine lunar samples and compared them with the commonly used $I_{s/FeO}$ parameter and C concentration, both of which can be used to represent surface maturity. The OMAT parameter is designed to minimize the spectral signature of composition in order to image surface maturity, but a small effect of FeO concentration may remain in this formulation. There is a good correlation between OMAT and $I_{s/FeO}$, especially if the Apollo 17 sample 71061, spectra of which contain a ~600 nm feature (likely an Fe^{2+}-Ti^{4+} charge transfer process, due to the presence of Fe- and Ti-bearing glass and oxide phases). Also shown is the correlation between optical maturity and continuum-removed depth of a spectral feature centred near 3 μm (Izawa et al., 2014, in press) that is related at least in part to hydration (hydroxyl groups and possibly molecular water) native to the lunar surface. If the anomalous sample 71061 is not considered, there is a general trend of decreasing OMAT (i.e. increasing surface maturity) with the depth of the 3 μm feature. To the extent that increasing band depth suggests increased concentration of the molecular group responsible, this may indicate increased hydration of samples with greater maturity.



Robinson et al., (2007) developed a technique for remote determination of lunar surface TiO_2 concentration using the ratio of reflectance at 502 nm to that at 250 nm in spectra from the Hubble Space Telescope. We have calculated the 502/250 nm reflectance ratio (A) and for comparative purposes, the 750/415 nm reflectance ratio, following Robinson et al., (2007). The highly scattered nature of both plots is ascribable to a combination of the large variation in maturity and lithology of our samples, and the existence of a ~600 nm feature in the spectra of sample 71061.

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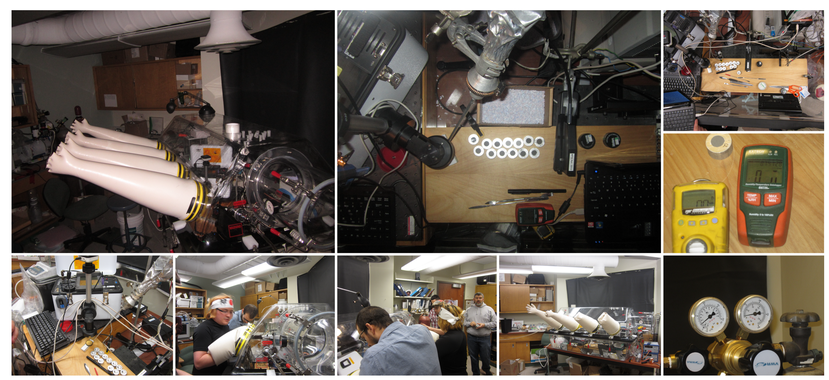
References: [1] E. A. Cloutis et al., Icarus 197, 321 (2008). [2] D. J. Vaughan, R. G. Burns, Phil. Trans. of the Royal Society A 285, 249 (1977). [3] D. E. Melendrez, J. R. Johnson, S. M. Larson, R. B. Singer, JGR -Planets 99, 5601 (1994). [4]J. R. Johnson, S. M. Larson, R. B. Singer, GRL 18, 2153 (1991). [5] P. G. Lucey, D. T. Blewett, B. L. Jolliff, JGR 105, 20297 (2000). [6] P. G. Lucey, D. T. Blewett, G. J. Taylor, B. R. Hawke, JGR 105, 20377 (2000). [7] P. G. Lucey, D. T. Blewett, B. R. Hawke, JGR 103, 3679 (1998). [8] P. G. Lucey, G. J. Taylor, E. Malaret, Science 268, 1150 (1995). [9] M. S. Robinson et al., GRL 34, (2007), [10] D. Blewett, P. G. Lucey, B. R. Hawke, B. L. Joliff, JGR 102, 16319 (1997).



Contamination of the Apollo samples with terrestrial oxygen, water, or other contaminants would compromise the goals of this study; therefore special procedures were implemented to preserve their integrity. Spectra were collected in a Plas-Labs 818 GBB glovebox with interior dimensions 97 × 152 × 79 cm; (Plas-labs Inc., Lansing, MI) under a dry N_2 atmosphere, where dry N_2 was passed through a series of Drierite filters. The glovebox also contained Drierite and Chemisorb to remove any remaining water and CO2 prior to opening of the sample containers. Humidity and O_2 levels were monitored continuously with a BW Technologies Gas Alert O_2 Extreme oxygen meter, and an Extech Instruments RH20 humidity/temperature datalogger, and were <<1% throughout the course of the measurements. Samples were only opened within the glovebox. Additional checks on sample integrity were provided by the inclusion of nanophase iron (50 nm nominal particle size) and calcium sulphide in the glove box, which are extremely sensitive to oxidation, hydration, or adsorption of water. Nanophase iron and CaS samples were monitored spectroscopically throughout the course of the experiment and by X-ray diffractometry (XRD) before and after and no evidence of oxidation, hydration or water adsorption within the glovebox environment was observed. For the XRD analysis, aliquots of nanophase iron and calcium sulphide were removed (through an independently purged airlock) from the glovebox at the beginning of the experiment and immediately analyzed using a Bruker D8 Discover X-ray diffractometer with Co $K\alpha_{1,2}$ radiation (λ=1.78897 Å) operating at 40 kV and 40 mA. After the entire experiment, aliquots of nanophase iron and calcium sulphide that had remained exposed within the glovebox for the duration were again removed and analyzed by XRD.

An important issue which is not within the control of the experimenters is that of pre-analysis contamination of the samples, either during their return to Earth or subsequently, either during the 40-43 years of curation or during shipment to the laboratory. The samples investigated in this study are splits of material that (to the best of our knowledge) were not exposed to air or water during either return or curation. The specific sample identities and parentage of the materials investigated here are summarized in Table 1. The samples were delivered to HOSERLab in sealed sample vials (packed under high-purity N_2 atmosphere at NASA Johnson Space Center) held in heavy plastic, within airtight steel containers further wrapped in heavy plastic, all held within a steel briefcase. There was no sign of damage to any of the packaging materials when they were removed from the case within the purged glovebox. We also opened the successive layers of packaging material with the O_2 and RH meters nearby, to observe any spike in either O_2 concentration or relative humidity at the moment of opening (which could indicate the presence of contamination), no such spike was observed.